

END-SEMESTER EXAMINATION, 2023-24
B. Tech. (SEMESTER : 05)
CSE 021 : INTRODUCTION TO CLOUD COMPUTING

Time: 3 Hrs.

Max. Marks: 100

- Note:** 1. All questions are compulsory.
 2. Assume missing data suitably, if any.

- CO1: Define the basics of cloud and recall the computer Science concepts which are helpful in understanding on demand service architecture.
 CO2: Classify and describe the architecture and taxonomy of parallel and distributed computing, including shared and distributed memory, and data and task parallel computing.
 CO3: Apply the PAAS and SAAS to manage the workflow and use of cloud in scientific application.
 CO4: Categorize and Characterize between Infrastructure services, deployment models, and governance in cloud computing. Examine the design of task and data parallel distributed algorithms for Clouds and use them to construct Cloud applications.
 CO5: Evaluate the importance of cloud using monitoring and management of services for performance improvement of HPC and to follow the Governance and Compliances.
 CO6: Elaborate the design concept and formulate to build the solution using cloud service providers as AWS, MS Azure, Google Cloud. Demonstrate the use of Map-Reduce, Vertex-Centric and Continuous Dataflow programming models.

COs Marks BTL

SECTION-A

All Questions are Compulsory:**(10×4=40 Marks)**

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|--|----------|
| 1. How does a Remote Procedure Call work? | CO1 4 K1 |
| 2. Explain the use of Web Sockets. | CO1 4 K2 |
| 3. Why are VMs crucial in cloud infrastructures? | CO2 4 K1 |
| 4. Relate the significance of VM Provisioning to the efficiency of cloud services. | CO2 4 K2 |
| 5. Explain RVWS design. | CO3 4 K2 |
| 6. Identify different advantages of using Aneka in cloud computing. | CO3 4 K3 |
| 7. Explain the automated policy based management. | CO4 4 K2 |
| 8. Construct the life cycle of SLA. | CO4 4 K3 |
| 9. Explain AWS services EC2 and Lambda. | CO5 4 K2 |
| 10. Identify and explain the basic requirements of a secure cloud software. | CO5 4 K3 |

SECTION-B

All Questions are Compulsory:**(3×6=18 Marks)**

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|--|----------|
| 11. (a) Analyse the key components of CometCloud's architecture. | CO3 6 K4 |
| --- OR --- | |
| (b) Examine Dynamic ICT services in detail. | |
| 12. (a) Classify PCI and DSS in cloud computing. | CO4 6 K4 |
| --- OR --- | |
| (b) Analyse game hosting on cloud resources. | |

13. (a) Discover the potential use cases for Azure's SQL Server on Virtual Machines. CO5 6 K4

--- OR ---

- (b) Explain the advantages of Cloud Storage. How to choose a Cloud Storage for business?

SECTION-C

All Questions are Compulsory:

(3×10=30 Marks)

14. (a) Categorise in detail about Aneka architecture. CO3 10 K4/K5

--- OR ---

- (b) Select and Explain the SAGA based scientific applications that utilize clouds.

15. (a) Discover the legal issues arising due to cloud computing, particularly related to data storage. CO4 10 K4/K5

--- OR ---

- (b) Examine the challenges associated with hosting games on cloud resources.

16. (a) Justify the integration of Azure Backup in a comprehensive disaster recovery plan. CO5 10 K4/K5

--- OR ---

- (b) Evaluate the importance of identity management using services like AWS's IAM and Azure's Active Directory.

SECTION-D

Attempt the following Question:

(1×12=12 Marks)

17. (a) Compare the different types of Google cloud platforms. CO6 12 K5/K6

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- (b) Criticize the architecture of Workflow Management Systems in relation to optimizing Scientific Applications for Cloud Environments.

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